

Chapter 26 — Infections of the Urinary Tract — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- Cystitis first-line: nitrofurantoin 100 mg twice daily × 5 d; TMP-SMX DS twice daily × 3 d (local resistance ≤20%); fosfomycin 3 g once; pivmecillinam
- Fluoroquinolones = last resort for cystitis (FDA 2016)
- Treat ASB only in pregnancy and before mucosa-breaching urological procedures
- Symptomatic cystitis culture threshold $\geq 10^2$ CFU/mL; Kass $\geq 10^5$ was for pyelonephritis
- Recurrent UTI = 2 in 6 months or ≥ 3 in 12 months; no routine imaging/cystoscopy if uncomplicated
- Vaginal (not oral) oestrogen prevents rUTI postmenopausally
- Pyelonephritis febrile >72 h → CT for obstruction/abscess
- Renal abscess ≤ 3 cm → antibiotics; >5 cm → percutaneous drainage; perinephric >3 cm → drain
- Infected obstructed kidney → stent or nephrostomy plus antibiotics
- AP: within 60 min (vancomycin/FQ 120 min); single dose default; ≤ 24 h max

DEFINITIONS AND EPIDEMIOLOGY

- **Complicated UTI:** abnormal tract, compromised host, or virulent/resistant organism; **male sex = complicated**
- **Box 26.1 flags:** pregnancy, elderly, DM, immunosuppression, catheter, instrumentation, hospital-acquired, **symptoms >7 d**
- **Reinfection** = new inoculation (same organism >2 weeks, or new organism); **persistence** = same organism from focus
- **Prophylaxis** presumes sterile tract; **suppression** targets ineradicable focus
- **CAUTI = commonest nosocomial infection** (about 80% of nosocomial UTI); **commonest cause of secondary bacteraemia**
- **Commonest:** *E. coli* (about 85% community, 50% hospital); *S. saprophyticus* about 10% cystitis in young women
- About 70% of recurrences = reinfection, different organism
- Uncomplicated rUTI → no scarring, hypertension or renal failure

PATHOGENESIS

- **Ascending route commonest;** haematogenous = *S. aureus* or *Candida*
- **Type 1 pili (FimH) → uroplakins → cystitis;** mannose-sensitive
- **P pili → glycolipid receptors → pyelonephritis;** mannose-resistant
- **IBC → fluxing → QIR** (quiescent reservoir) drives recurrence
- **Adult VUR harmless** unless high pressure/stasis + UTI
- **Papillary necrosis:** DM, analgesics, sickle cell, obstruction
- **HIV/AIDS:** UTI about 5×; **CD4 <200 → TMP-SMX** (also lowers UTI)

EVALUATION, IMAGING, MICROBIOME

- **Nitrite:** specific, insensitive; **all Enterobacteriaceae +;** Gram-positives, *Pseudomonas* –
- **Best dipstick combination:** nitrite, or leucocyte esterase + blood
- **Pyuria:** ≥ 5 –10 WBC/HPF; **>50 + symptoms justifies antibiotics**
- **Sterile pyuria → TB, stones, urothelial injury, interstitial nephritis**
- **Culture = gold standard;** commensals: CoNS, diphtheroids, α /non-haemolytic streptococci
- **Culture +, no pyuria → catheterised sample**
- **Catheter patient:** replace catheter before sampling; **stoma:** catheterise stoma
- **Gram-positives in voided sample rarely cause cystitis alone**
- **Painless haematuria → full urological evaluation**
- **Stone culture > urine culture predicts post-PCNL sepsis**
- **Non-contrast CT = stones; pelvic MRI = urethral diverticulum**
- **Box 26.5:** image pyelonephritis if **no response after 5–6 d**

ASYMPTOMATIC BACTERIURIA

- **Women: 2 voided $\geq 10^5$; men: 1 voided $\geq 10^5$; catheter: $\geq 10^2$ CFU/mL**
- **Long-term catheter ASB = 100%;** long-term care 25–50% women
- **Don't treat:** DM, elderly, SCI, catheters, children, **renal transplant >1 month**, non-urological surgery; **pyuria not indication**
- **Neutropenia:** no consensus
- **Cai trial:** treating ASB in rUTI ↑ **recurrence** (about 70% vs 38%) and resistance
- **Arthroplasty:** ASB marks risk; **treatment doesn't help**

ANTIMICROBIALS

- **Nitrofurantoin:** no tissue levels → **never pyelonephritis;** CrCl cut-off <30 (Beers 2015); G6PD haemolysis; pulmonary fibrosis
- **TMP-SMX:** misses *Enterococcus*, *Pseudomonas*; **commonest antibiotic cause of SJS/TEN**
- **Fosfomycin:** single dose; covers VRE, MDR; low cross-resistance
- **FQ adjust CrCl <30; FDA warnings:** tendon rupture, neuropathy, FQAD, CNS, dysglycaemia, **aortic dissection**
- **Ceftriaxone misses *Pseudomonas*; ceftazidime covers**
- **Cephalosporins never cover enterococci**
- **Carbapenem gaps:** *E. faecium*, MRSA; ertapenem also no *Pseudomonas/Enterococcus*
- **CDI:** ≥ 3 unformed stools/24 h + toxin; **fidaxomicin first line;** vancomycin alternative; **FMT after ≥ 2 recurrences;** fulminant → oral vancomycin 500 mg + IV metronidazole
- **CDI culprits:** cephalosporins, clindamycin, carbapenems, FQ, TMP-SMX

BLADDER INFECTIONS

- **Cystitis duration:** 3 d standard; nitrofurantoin 5 d; fosfomycin single
- **Afebrile men: 7 d = 14 d** (Drekonja 2021)
- **Complicated UTI:** culture mandatory; **10–14 d;** correct obstruction
- **Emphysematous cystitis:** elderly diabetic women; **gas in bladder wall; 90% medical**
- **Unresolved bacteriuria — commonest cause:** pre-existing resistance
- **Still-sensitive organism on treatment → investigate** (azotaemia, papillary necrosis, staghorn)
- **Rapid reinfection mimicking unresolved → enterovesical fistula**
- **Struvite:** urease (*Proteus*); **remove if fit;** else acetohydroxamic acid + suppression

- Observe asymptomatic non-obstructing stones ≤ 10 mm
- Water ≥ 1.5 L/day halves cystitis (premenopausal)
- Cranberry ≥ 36 mg PAC twice daily; safe with warfarin
- Raz-Stamm 1993: vaginal oestriol cut UTIs about 90%; benefit ≥ 12 weeks
- Methenamine hippurate 1 g twice daily: needs acid urine; CrCl < 50 contraindicated; useless with urease-splitters; = antibiotic prophylaxis (RCT)
- Postcoital > daily prophylaxis for coital rUTI

KIDNEY INFECTIONS

- Outpatient pyelonephritis: ciprofloxacin 500 mg twice daily $\times 7$ d if FQ resistance $\leq 10\%$; levofloxacin 750 mg $\times 5$ d; TMP-SMX $\times 14$ d only if susceptible
- Complicated: 2–3 d IV if blood cultures negative, then oral 10–14 d
- Solitary obstructed kidney $\rightarrow$ acute renal failure; dose by GFR
- Emphysematous pyelonephritis: diabetic women; CT of choice; streaky gas, no fluid $\rightarrow$ 50–60% mortality; fluid/loculated gas $< 20\%$
- EPN management: drain; nephrectomy for diffuse gas with destruction
- Emphysematous pyelitis = collecting-system gas; benign
- Renal abscess: now Gram-negative ascending; haematogenous $\rightarrow$ antistaphylococcal penicillin (vancomycin if allergic)
- Perinephric abscess: symptoms > 5 d, fever > 4 d on antibiotics; aminoglycoside + antistaphylococcal
- XGP: stones (staghorn) + non-function; *Proteus*; CT triad; nephrectomy (partial if focal)
- Malakoplakia: Michaelis–Gutmann bodies, von Hanseman cells; women 4:1; $\alpha 1$ -antitrypsin stain; intracellular antibiotics (FQ, TMP, rifampicin, doxycycline); bilateral/transplant $\rightarrow$ death within 6 months

OTHER INFECTIONS

- Fournier's: source found about 95%; men 10:1; Buck's $\rightarrow$ dartos $\rightarrow$ Colles' $\rightarrow$ Scarpa's
- LRINEC ≥ 6 suspicious, ≥ 8 strongly predictive
- CT most specific; debride to normal fascia, re-look 24–48 h; orchidectomy rarely; mortality 20–40%

PREGNANCY AND ELDERLY

- Screen by culture at 12–16 weeks (or first visit); dipstick misses
- Untreated bacteriuria $\rightarrow$ pyelonephritis (historically 15–45%); 60–75% third trimester

- GFR +30–50%; creatinine 0.5, urea 9 mg/dL; proteinuria abnormal > 300 mg/24 h
- Safe: penicillins, cephalosporins; avoid: FQ, tetracycline (D), chloramphenicol (grey baby), TMP first trimester, nitrofurantoin at term
- After 1 pyelonephritis or 2 cystitis $\rightarrow$ nitrofurantoin/cephalexin suppression
- Pregnant pyelonephritis: admit, IV β -lactam; ultrasound if no response 72 h
- Elderly: dysuria key; confusion alone $\neq$ UTI $\rightarrow$ hydrate, reassess; never culture for clearance; short (3–6 d) = long course

CATHETERS, SCI, TRANSPLANT

- CAUTI: catheter > 2 d + symptom + $\geq 10^5$; bacteriuria 5–10%/day
- Catheter significant bacteriuria $\geq 10^2$; pyuria non-discriminatory
- Treat only symptomatic CAUTI; change catheter; meatal care with water only
- SCI: CIC lowest long-term risk; no screening/treating ASB; no AP
- SCI significant: indwelling any count; CIC $\geq 10^2$; voided men $\geq 10^4$
- SCI duration: 5–7 d new afebrile, 7–10 d recurrent, 14 d febrile
- Catheter > 10 years $\rightarrow$ bladder cancer risk
- Transplant: uncomplicated 5–10 d; pyelonephritis 14–21 d; stents out within 4 weeks

ANTIMICROBIAL PROPHYLAXIS

- Diagnostic cystoscopy: no AP (EAU strongly against)
- TURP: AP all; operating on infected urine $\rightarrow$ bacteraemia in half
- SWL: no AP if urine benign; infection stones $\rightarrow$ full course first
- PCNL, prostheses: ≤ 24 h; ureteroscopy single oral dose
- AP at stent/catheter removal; urine culture poorly predicts stent organism
- TRUS biopsy: FQ-resistant *E. coli* commonest cause; antibiotics within 6 months = main risk; swap FQ if local resistance $> 20\%$; ≥ 1 day
- UDS AP: neurogenic LUTD, raised PVR, ASB, immunosuppression, age > 70 , catheters, joint < 2 years
- Endocarditis: AP not recommended; high-risk + infected urine $\rightarrow$ amoxicillin/ampicillin
- Joint > 2 years, pins/plates/screws $\rightarrow$ no AP; < 2 years $\rightarrow$ FQ or ampicillin + gentamicin
- Routine vancomycin AP not recommended

CLASSIC TRAPS

- Trap: pyuria + bacteriuria = UTI — **Actually:** symptoms decide; treat neither alone
- Trap: $\geq 10^5$ needed for cystitis — **Actually:** $\geq 10^2$ CFU/mL with dysuria
- Trap: nitrofurantoin for early pyelonephritis — **Actually:** no tissue levels; contraindicated
- Trap: confused elderly + positive culture = UTI — **Actually:** hydrate and reassess
- Trap: fever persisting on day 2 of pyelonephritis = failure — **Actually:** normal to 72 h; image after
- Trap: negative urine excludes pyonephrosis/abscess — **Actually:** complete obstruction or haematogenous source gives sterile urine
- Trap: oral oestrogen prevents rUTI — **Actually:** only vaginal oestrogen works
- Trap: treat ASB before arthroplasty — **Actually:** no benefit; don't screen
- Trap: cephalosporin covers enterococcus — **Actually:** never; use ampicillin
- Trap: XGP on frozen section = clear cell RCC — **Actually:** foamy macrophages mimic it; remove whole mass